Thirty pigeonhole problems

Mathematics, discrete mathematics, and theoretical computer science

Write $[n] = \{1, \dots, n\}$. All graphs are simple unless stated otherwise.

1. Two vertices with the same degree

Prove that every simple graph on $n \geq 2$ vertices contains two vertices of the same degree.

2. One chosen number divides another

Choose $n+1$ distinct integers from $\{1, 2, \dots, 2n\}$. Prove that two chosen integers $a < b$ satisfy $a \mid b$. Show that $n+1$ cannot be replaced by n.

3. A consecutive block divisible by n

Given arbitrary integers $a_1, \dots, a_n$, prove that some nonempty consecutive block has sum divisible by n:

$$n \mid (a_{i+1} + \cdots + a_j) \qquad (0 \leq i < j \leq n).$$

Negative entries are allowed; no positivity assumption may be used.

4. Basic Diophantine approximation

(a) Given $\alpha \in \mathbb{R}$ and an integer $N \geq 1$, prove that there are $p \in \mathbb{Z}$ and $1 \leq q \leq N$ with

$$|q\alpha - p| < \frac{1}{N}.$$

(b) If α is irrational, deduce that infinitely many distinct rational numbers p/q satisfy

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^2}.$$

(c) Generalize (a): for $\alpha_1, \dots, \alpha_d \in \mathbb{R}$ and an integer $Q \geq 1$, find integers $1 \leq q \leq Q^d$ and $p_1, \dots, p_d$ such that

$$\max_{1 \leq j \leq d} |q\alpha_j - p_j| < \frac{1}{Q}.$$

5. Fibonacci residues

Let $F_0 = 0$, $F_1 = 1$, and $F_{k+2} = F_{k+1} + F_k$. Prove that for every integer $m \geq 2$, there is a t with $1 \leq t \leq m^2$ such that

$$F_t \equiv 0 \pmod{m}, \qquad F_{t+1} \equiv 1 \pmod{m}.$$

Deduce that $F_{k+t} \equiv F_k \pmod{m}$ for every $k \geq 0$, and hence some positive-index Fibonacci number is divisible by m.

Problems 6–10

6. Integer points in the plane

Let $S \subseteq \mathbb{R}^2$ be a bounded measurable set of area greater than 1. Prove that it contains distinct points x, y with $x - y \in \mathbb{Z}^2$.

Deduce the planar form of Minkowski's theorem: if a bounded convex set $K \subseteq \mathbb{R}^2$ is centrally symmetric about the origin and has area greater than 4, then K contains a nonzero integer point. No regularity of the boundary is assumed.

7. Two disjoint selections with equal sum

Let $a_1, \dots, a_n$ be positive integers, each at most M. Assume

$$2^n > nM + 1.$$

Prove that there are disjoint nonempty index sets $A, B \subseteq [n]$ with

$$\sum_{i \in A} a_i = \sum_{i \in B} a_i.$$

The integers a_i need not be distinct.

8. How large can an intersecting family be?

Let $n \geq 1$, and let $\mathcal{F}$ be a family of subsets of $[n]$ such that any two distinct members intersect. Prove that

$$|\mathcal{F}| \leq 2^{n-1}.$$

Give a family attaining equality. The sets in the family may have different sizes.

9. A long monotone subsequence

For positive integers r, s, prove that every sequence of $rs + 1$ distinct real numbers has either a strictly increasing subsequence of length $r + 1$ or a strictly decreasing subsequence of length $s + 1$. A subsequence need not be consecutive.

Construct a sequence of length rs showing that the threshold is sharp.

10. Six people

Among six people, every pair are either acquainted with each other or strangers to each other; acquaintance is symmetric. Prove that there are either three people who are pairwise acquainted or three people who are pairwise strangers.

Show that the conclusion need not hold for a group of five people.

Problems 11–15

11. Monochromatic rectangles, with a sharp example

(a) Every entry of a 3×7 array is red or blue. Prove that some two distinct rows and two distinct columns have all four corner entries of the same color. The rows and columns need not be consecutive.

(b) Construct a 3×6 array with no such rectangle.

(c) Generalize: prove that every r-coloring of a

$$(r+1) \times \left(r \binom{r+1}{2} + 1 \right)$$

array has a monochromatic rectangle. No sharpness claim is requested for general r.

12. Hamiltonian cycles

Let G be a simple graph on $n \geq 3$ vertices with minimum degree at least $n/2$. Prove that G has a Hamiltonian cycle.

13. Antichains

A family $\mathcal{A} \subseteq 2^{[n]}$ is an *antichain* if no distinct member contains another. Prove the stronger inequality

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq 1,$$

and deduce $|\mathcal{A}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

14. Bipartite matchings

In a bipartite graph with parts L, R, write $N(S) \subseteq R$ for the neighbors of $S \subseteq L$. Prove that a matching covering L exists if and only if

$$|N(S)| \geq |S| \quad \text{for every } S \subseteq L.$$

Also give an example showing that $|L| \leq |R|$ alone is insufficient.

15. Binary codes

Let $0 \leq t \leq n$, and let $C \subseteq \{0,1\}^n$ be a binary code whose distinct words have Hamming distance at least $2t+1$. The Hamming distance is the number of coordinates in which two words differ. Prove

$$|C| \sum_{j=0}^{t} \binom{n}{j} \leq 2^n.$$

For $n = 7$ and $t = 1$, deduce $|C| \leq 16$.

Problems 16–20

16. Boolean circuits

A circuit has $n \geq 1$ Boolean inputs and one designated output wire. Each gate may compute any of the 16 Boolean functions of two inputs; both incoming wires come from input variables or earlier gates. Repeated wires are allowed, and circuit size counts gates.

Prove that for some absolute $c > 0$ and all sufficiently large n, a Boolean function on n inputs requires more than $c2^n/n$ gates. Prove also that the fraction of Boolean functions computable with at most $\lfloor 2^n/(16n) \rfloor$ gates tends to zero as $n \to \infty$.

17. Remembering the nth symbol from the end

Fix $n \geq 1$. A deterministic finite automaton must recognize exactly the binary strings of length at least n whose nth symbol from the end is 1. Prove that at least 2^n states are necessary, and construct an automaton with exactly 2^n states.

18. Testing equality

Alice receives $x \in \{0,1\}^n$ and Bob receives $y \in \{0,1\}^n$, where $n \geq 1$. They use a deterministic binary communication protocol. At every node of its protocol tree, the speaker is fixed and the next bit depends only on that speaker's input and the transcript so far. Leaves carry the common output, which must be 1 exactly when $x = y$.

Prove that the exact minimum worst-case communication is $n + 1$ bits.

19. Finding a collision

Let $N \geq 2$. An unknown function $f : [N+1] \to [N]$ is accessible only by queries that reveal $f(i)$. A deterministic algorithm must always output distinct i, j with $f(i) = f(j)$.

Prove that the exact worst-case query complexity is $N + 1$. The algorithm may query adaptively and may even output an index it has not queried.

20. Random collisions

Let $X_1, \ldots, X_k$ be independent uniform samples from $[M]$, where $2 \leq k \leq M$. Prove

$$1 - e^{-k(k-1)/(2M)} \;\leq\; \Pr[\exists i < j : X_i = X_j] \;\leq\; \frac{k(k-1)}{2M}.$$

Deduce that a constant collision probability appears with $\Theta(\sqrt{M})$ samples, although a collision is guaranteed for arbitrary samples only after $M + 1$ samples.

Problems 21–25

21. Five points in a square

Five distinct points lie in a square of side length 1, with boundary points allowed. Prove that two of the points are at distance at most $1/\sqrt{2}$.

22. Name cards around a table

There are $n \geq 2$ equally spaced seats around a circular table, with one person and one name card at each seat. The cards contain the names of the n people, each exactly once. Initially no person is sitting in front of their own name card.

Prove that rotating all the cards together through a suitable whole number of seat positions puts at least two people in front of their own name cards.

23. Divisible selections

Prove that among any seven integers one can select four whose sum is divisible by 4.

More generally, prove that for every integer $k \geq 1$, among any $2^{k+1} - 1$ integers one can select 2^k whose sum is divisible by 2^k. Repeated values are allowed, and selections use distinct positions in the given list.

24. A square product

Seventeen distinct positive integers are given. Every prime divisor of each of them belongs to $\{2, 3, 5, 7\}$. Prove that the product of two of the given integers is a perfect square.

25. A colored plane

Every point of the Euclidean plane is colored either red or blue. Prove that there is a nondegenerate equilateral triangle whose three vertices have the same color.

Problems 26–30

26. Numbers around a circle

The numbers $1, 2, \ldots, 9$ are placed around a circle in an arbitrary order, each exactly once. Prove that three consecutive numbers have sum at least 16. Consecutive triples may cross the end of any chosen written listing.

27. Greeting cards

Twenty children each send a greeting card to ten different other children. Prove that two children send cards to each other.

28. A tournament

In a tournament, every pair of players plays exactly one game and no game is drawn. Prove that there is a player P with the following property: for every other player Q, either P defeats Q, or P defeats a player who defeats Q.

29. Nearest neighbors

Let S be a finite set of at least two distinct points in the Euclidean plane. Suppose every point of S has a unique nearest neighbor in S. Prove that no point of S is the nearest neighbor of more than five other points.

30. An infinite tree

An infinite rooted tree has only finitely many children at each vertex. Prove that there is an infinite sequence of distinct vertices $v_0, v_1, v_2, \ldots$ such that v_0 is the root and v_{i+1} is a child of v_i for every $i \geq 0$.